



Ethical Issues in Automated Hiring Systems: The Case of HireVue

TECHNICAL AND ETHICAL ANALYSIS OF AI-ASSISTED RECRUITMENT

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Introduction

Automated hiring refers to the use of algorithms, artificial intelligence, and other automated technologies to support stages of employee recruitment. These systems can screen applications, assess candidates, rank applicants, and provide information to human recruiters before a final employment decision is made. Organisations are attracted to them because they may process large applicant pools more efficiently, reduce recruitment costs, and provide greater consistency than fully manual processes. Raghavan et al. (2020) describe algorithmic pre-employment assessments as tools for evaluating candidates before later recruitment stages, while Köchling and Wehner (2020) identify efficiency, standardisation, and reduced human subjectivity as important reasons for adopting algorithmic HR systems.

These potential benefits have encouraged organisations to use automated assessment in real recruitment settings. HireVue is one example. Harwell's (2019) investigation for The Washington Post reported that HireVue's video-assessment system could analyse information including facial movements, word choices, and speaking voices and use these features to support candidate assessment and ranking. Employers were attracted to such systems partly because they could reduce the time and human resources needed to process large applicant pools. Ideally, an automated hiring system should help employers identify candidates whose abilities are relevant to a particular role while reducing unnecessary workload for HR professionals.

Recruitment decisions directly affect applicants' access to employment opportunities. The value of automated hiring therefore depends not only on efficiency, but also on whether the system uses valid job-related indicators, treats applicants fairly, and provides sufficient transparency about how assessments are produced. If the indicators do not reliably represent future job performance, suitable candidates may be screened out while less suitable candidates progress further. Limited transparency can also make it difficult for applicants and employers to understand, question, or identify possible errors. These concerns make validity, fairness, and transparency important professional issues rather than purely technical questions.

This report uses HireVue as its primary case study and the iTutorGroup incident as a supporting example. The U.S. Equal Employment Opportunity Commission (EEOC, 2023) reported that iTutorGroup agreed to settle a case involving allegations that its application software automatically rejected female applicants aged 55 or older and male applicants aged 60 or older. The report investigates the question: How can technical limitations in automated hiring systems create ethical risks for job applicants, employers, and computing professionals? It first examines how automated hiring and HireVue's assessment technology operate, then reviews research on validity, bias, transparency, and applicant impact. The Problem Analysis examines how technical limitations may lead to ethical consequences and considers the responsibilities of the stakeholders involved.

Literature Review

2.1 AUTOMATED HIRING SYSTEMS

Automated hiring systems support applicant screening, assessment, ranking, and candidate selection. Raghavan et al. (2020) examined algorithmic pre-employment assessment vendors and found that such systems use different forms of candidate information to predict employment-related outcomes. Employers and developers must therefore decide both what characteristics to measure and what outcome the model should predict. Success might be represented by job performance, employee retention, sales outcomes, performance reviews, or another indicator.

Köchling and Wehner (2020) place these systems within the broader development of algorithmic HR decision-making. Their systematic review shows that organisations are interested in automation because it may increase efficiency and consistency while reducing some forms of human subjectivity. Their review also identifies concerns about discrimination, fairness, and applicant perceptions. Automated hiring does not simply replace a recruiter with a machine: employers still decide how the system is introduced, what information matters, and how much influence an automated assessment has over later decisions.

Woods et al. (2020) similarly argue that digital selection changes the relationship between selection validity and applicant reactions. A process may be efficient and standardised yet still be experienced negatively by candidates or fail to predict relevant performance. Taken together, these studies suggest that automation alone does not guarantee a better hiring process. Its benefits depend on what the system measures, how it is designed, and how employers use the results.

2.2 HIREVUE AND AUTOMATED VIDEO ASSESSMENT

HireVue provides a specific example of how automated assessment has been applied to recruitment. Sánchez-Monedero et al. (2020) examined HireVue alongside other automated hiring systems and described the use of video interviews and other assessments before later recruitment stages. During the period covered by their research, HireVue could process categorical, audio, and video indicators and use these inputs to estimate an applicant's suitability for a particular role.

Harwell's (2019) investigation provides further information about how the technology was used at that time. Candidates could complete recorded interviews using a computer or mobile device, while the system analysed facial movements, word choice, and speaking voice. These features could contribute to assessments of competencies and behaviours, which were then used to support applicant ranking. Human hiring managers retained decision-making authority, although the automated assessment could influence which applicants received further attention.

The assessment process also used information from employees already working in the relevant role. According to HireVue representatives interviewed by Harwell (2019), existing employees could complete similar assessments and their responses could be compared with measures such as sales performance or customer-service outcomes. Patterns

associated with stronger-performing employees could then be used when evaluating new applicants. This is an application of machine learning: patterns identified from existing data are used to make predictions about new candidates.

Sánchez-Monedero et al. (2020) did not have access to HireVue's complete source code, datasets, or trained models. Their description therefore provides useful information about publicly documented operation but cannot independently verify every feature or model used in practice. This limitation does not make the source useless; it sets a boundary around what the study can establish. Its findings should be interpreted as evidence about documented claims and practices, rather than as a complete technical audit of HireVue.

2.3 VALIDITY, DATA SELECTION AND BIAS

A major concern in automated hiring research is whether the indicators used by a system provide a valid basis for predicting future job performance. Raghavan et al. (2020) explain that developers and employers must first convert an abstract idea such as a “good employee” into measurable outcomes. A measurable correlation between an applicant characteristic and an employment outcome does not automatically establish that the characteristic is a meaningful or appropriate indicator of job suitability.

This issue is particularly relevant to assessments that use behavioural information. Sánchez-Monedero et al. (2020) describe how HireVue could use audio and video indicators, including characteristics of how applicants spoke, as part of its assessment process. They also describe HireVue’s attempts to identify and reduce adverse impact by removing or adjusting features associated with demographic differences. However, publicly available information did not provide enough detail to independently evaluate every feature or bias-mitigation method used by the system.

Data selection creates another challenge. Raghavan et al. (2020) note that employers often possess detailed performance information only for people who were previously hired. They do not know how rejected applicants would have performed if employed. Models trained on existing employees may therefore learn patterns from a population already shaped by earlier human hiring decisions. Köchling and Wehner (2020) identify related forms of bias, including historical, representation, technical, and emergent bias. Köchling et al. (2021) further show why predictive accuracy and fairness must be evaluated separately in algorithmic video analysis. Historical bias can arise when existing data reflects earlier inequalities; representation bias can occur when groups are insufficiently represented; and technical bias can emerge when complex human characteristics are converted into simplified variables.

The evidence base also has limitations. Köchling and Wehner’s (2020) systematic review offers broad coverage, but much of the literature available at that time was non-empirical. Raghavan et al. (2020) examined many vendors but relied heavily on information made public by companies themselves. The connection between these limitations matters: if behavioural indicators, performance targets, or reference data are poorly chosen, bias can enter before the model makes a prediction. Later bias-mitigation efforts may then be trying to correct a problem embedded in the definition of success rather than only in the final output.

2.4 HIREVUE CONTROVERSY, TRANSPARENCY AND RESPONSE

Concerns about HireVue became more visible in 2019 as applicants, researchers, and privacy organisations questioned how automated video assessments evaluated candidates. Transparency became important because applicants could receive an assessment outcome without fully understanding which features influenced their score or how those features were interpreted.

In November 2019, the Electronic Privacy Information Center (EPIC, 2019) filed a complaint with the U.S. Federal Trade Commission. EPIC raised concerns about HireVue's use of facial analysis, applicant privacy, algorithmic opacity, and the ability of candidates to understand how they were assessed. The complaint is evidence of concerns raised by an advocacy organisation, not a final legal ruling. Its claims should therefore be presented as allegations rather than proof that HireVue had legally violated the standards described by EPIC.

HireVue later introduced changes to its assessment practices. Zuloaga (2021) reported for HireVue that the company had removed visual analysis from new pre-hire assessment models. HireVue stated that its internal research found visual information contributed less predictive value than language-based information, while improvements in natural language processing increased the usefulness of linguistic features. The company also published information about an algorithmic audit conducted by O'Neil Risk Consulting & Algorithmic Auditing, presenting the audit and removal of visual analysis as part of its approach to responsible and transparent AI.

External analysis examined the scope of this audit more cautiously. Engler (2021), writing for the Brookings Institution, noted that the audit examined documentation for one assessment and did not independently analyse the underlying data or directly test all of HireVue's models. The three perspectives therefore use different standards when discussing transparency. HireVue focuses on auditing and changes to its technology; EPIC focuses more strongly on whether applicants can understand and challenge automated decisions; Brookings considers how much an audit can actually verify. Technical changes and auditing can improve transparency without necessarily resolving every transparency concern.

2.5 APPLICANT EXPERIENCE AND THE ITUTORGROUP SUPPORTING EXAMPLE

The effects of automated hiring are also experienced directly by applicants. Langer et al. (2020) found that participants could view highly automated interviews as more consistent while also experiencing lower social presence. Lower social presence was associated with less positive perceptions of the organisation. Because this research did not specifically test HireVue, it should not be treated as direct evidence of HireVue applicants' reactions. Instead, it provides broader empirical evidence that automation can change how applicants experience and evaluate a recruitment process.

A different form of automated hiring problem can be seen in the iTutorGroup case. The U.S. Equal Employment Opportunity Commission alleged that iTutorGroup had programmed its application software to automatically reject female applicants aged 55 or

older and male applicants aged 60 or older. According to the EEOC, more than 200 qualified applicants were affected. The dispute was later resolved through a US\$365,000 settlement and a consent decree involving training, anti-discrimination policies, and monitoring of future hiring practices (EEOC, 2023).

Wiessner (2023), reporting for Reuters, recorded that iTutorGroup denied wrongdoing. This distinction matters because a settlement does not itself mean that every allegation was established by a final court judgment. EEOC material is useful for documenting the regulator's allegations and settlement terms, while Reuters provides an additional journalistic account that includes the company's position.

The technical mechanism in iTutorGroup also differs from HireVue. HireVue involved machine-learning-based assessment of candidate information, while iTutorGroup involved a more direct automated screening rule. Although these systems used different forms of automation, both could influence whether applicants were allowed to continue through recruitment. They therefore belong in the same ethical discussion because different technical mechanisms can affect the same high-stakes outcome: access to an employment opportunity.

Problem Analysis: How, What, Why

3.1 HOW THE SYSTEM SHOULD WORK

An automated hiring system should support, rather than replace, a fair and job-related recruitment process. Its expected benefits are practical: processing a large applicant pool efficiently, reducing administrative cost, simplifying early-stage screening, and reducing repetitive work for HR staff. It should also help employers identify candidates whose skills and behaviours are relevant to the specific role. A system that simply produces a fast score is not meeting its purpose if the score is unrelated to the work or if employers cannot interpret its limitations.

The ideal process would begin with a clearly defined performance target, use indicators that have a defensible relationship with that target, and test those indicators across relevant applicant groups. The system would be monitored after deployment, with human reviewers able to question or override an output. Applicants and employers would also receive enough information to understand what the system is designed to do and what it cannot establish. The comparison with iTutorGroup reinforces that automation does not remove the employer's responsibility for the decision process: a programmed rule can be simple and still create serious employment consequences.

3.2 WHAT WENT WRONG: VALIDITY, ACCESSIBILITY AND CLASSIFICATION ERROR

The first problem is validity. HireVue's earlier assessment practices used facial movement, voice, language, and other behavioural information as inputs to a prediction about candidate suitability. These features can be measured, but measurement does not prove that they are valid indicators of future job performance. A speaking pace, facial expression, accent, or style of language may reflect culture, disability, communication context, or interview anxiety rather than the competency needed for a role. If the

relationship between an indicator and performance is weak, the system can misclassify applicants.

The practical consequences are two-sided. A false negative occurs when a suitable candidate is screened out; a false positive occurs when a less suitable candidate progresses and may later create additional recruitment, training, or replacement costs. In both cases, the system undermines its original efficiency objective. HireVue's later decision to remove visual analysis, as reported by Zuloaga (2021), is not proof that every earlier assessment was invalid, but it is relevant evidence that the company no longer considered visual information to add enough predictive value to new models.

Validity must also be considered across different applicants. Fisher et al. (2024) explain that AI-based selection tools can create barriers for applicants with disabilities when reliability, validity, accessibility, and applicant experience are not considered during design. The ethical issue is not that disabled applicants are inherently unsuitable. It is that a selection tool may measure a mode of expression or interaction that is not necessary for the job and may not work equally well for all candidates. A system can therefore appear consistent while still producing an unfair or inappropriate assessment.

The iTutorGroup example shows a more explicit classification problem. If software applies age-based thresholds to application data, it can reject applicants before a human assesses their skills or experience. The case was described by the EEOC as a legal allegation and was later settled; it should not be rewritten as a final finding of intentional discrimination. Nevertheless, it demonstrates why a technical decision rule must be examined in relation to the protected characteristics it uses and the employment opportunity it controls.

3.3 WHY THE PROBLEM CAN OCCUR: DATA, TARGETS AND SOCIOTECHNICAL CONTEXT

A second problem arises before the model produces an output. To train or calibrate an assessment, organisations must decide what counts as success, which data represents successful employees, and which features are relevant. If behavioural features and performance targets are derived mainly from existing employees, the system may learn from a workforce shaped by earlier human decisions. It may then favour applicants who resemble past employees even when those patterns reflect historical exclusion rather than genuine job requirements. This is how historical and representation bias can be reproduced or reinforced.

The danger is not limited to intentionally discriminatory design. A model can be technically accurate in reproducing a historical pattern while remaining ethically inappropriate. Selbst et al. (2019) describe fairness as a problem of sociotechnical systems rather than isolated models. Developers, employers, applicants, organisational practices, data collection, and social context all affect the meaning of an apparently neutral prediction. For automated hiring, this means that improving the model's statistical performance cannot by itself resolve a problem created by the choice of target or the way the result is used.

Drage and Mackereth (2022) add a related caution about claims that AI can “eradicate difference” by removing race or gender variables. Removing a protected characteristic does not necessarily remove the social patterns represented in other variables, nor does it show that the chosen performance target is fair. A system can therefore pass a narrow test while leaving deeper structural problems untouched. The relevant professional question is not only whether a variable is present, but what the system is measuring, why that measure is appropriate, and what groups may be disadvantaged by it.

Leicht-Deobald et al. (2019) also warn that algorithm-based HR decisions can encourage excessive trust in quantitative indicators. When employers treat an automated score as objective, they may fail to question whether it is job-related or whether it should be decisive in a particular case. This creates a potential feedback loop in practice: an organisation trusts the score, selects candidates who fit the score, treats their later performance as confirmation, and then continues to rely on the same standard. Even without automatic retraining, the process can reinforce an existing pattern.

3.4 WHY WE CARE: TRANSPARENCY, ACCOUNTABILITY AND SCALE

Transparency does not necessarily create the original validity or bias problem, but it affects whether the problem can be found and corrected. If an applicant cannot see what influenced a score and HR cannot explain which indicators mattered, it becomes difficult to distinguish a job-related assessment from an unreliable or biased indicator. The applicant may lose trust in the recruitment process, while HR may lose the ability to explain or review its own selection standard. Errors are then harder to challenge, investigate, and remedy.

Tabassi (2023) presents accountability and transparency as parts of trustworthy AI risk management and connects transparency with design decisions, training data, intended use, deployment, and the ability to support redress. This is relevant to HireVue because transparency should not be reduced to a public statement or a single audit. It should allow the relevant people to understand enough about the system’s purpose, limits, monitoring, and decision pathway to respond when harm occurs.

Responsibility should therefore be shared, but not vague. Developers are responsible for selecting and validating indicators, testing performance and adverse impact, documenting limitations, monitoring deployed systems, and communicating what the tool can and cannot establish. Employers are responsible for deciding whether the tool is suitable for the role, avoiding blind reliance on a score, maintaining meaningful human review, and taking responsibility for the final employment decision. If a developer has disclosed clear limitations but an employer still treats the score as decisive, employer responsibility becomes more significant. If the design is opaque and the employer cannot reasonably understand the limitations, developer responsibility becomes more significant. The responsibilities differ, but automation does not make either party disappear.

Finally, a systematic weakness can become a scalable problem. A single unreliable indicator may affect one decision; a widely deployed hiring system can repeat the same weakness across many applicants and organisations. The iTutorGroup settlement shows that even a simple automated rule can affect a large group and trigger policy, training, and

monitoring requirements. For computing professionals, the central concern is therefore not only whether a system works in a laboratory. It is whether its assumptions remain defensible when embedded in real organisations, used by people with different levels of understanding, and applied repeatedly to decisions that shape access to work.

Conclusion and Discussion

This report examined how technical limitations in automated hiring systems can create ethical risks for applicants, employers, and computing professionals. HireVue was introduced as a system intended to make recruitment more efficient, consistent, and scalable. Its earlier use of behavioural, audio, and visual information shows, however, that a measurable feature is not automatically a valid indicator of job performance. If the feature, performance target, or reference data is poorly chosen, the system may generate false negatives and false positives while appearing objective.

The analysis also showed that bias can enter before prediction through the selection of data, targets, and features. Historical patterns in existing employees can be reproduced or reinforced, while a narrow audit or technical change cannot by itself resolve every transparency concern. Applicant experience and accessibility matter as well: a system may be consistent but still reduce social presence, create barriers for applicants with disabilities, or make it difficult for applicants to understand and challenge an outcome. The iTutorGroup supporting example demonstrates that the risk is not limited to complex machine-learning systems; a direct automated rule can also remove access to employment opportunities.

Computing professionals need to care because their responsibility extends beyond making an automated system run. They must ask what the system measures, whether the measure is related to the job, how it performs for different groups, how its limitations are communicated, and how errors will be detected and corrected. Employers must retain meaningful oversight and responsibility for the final decision. When a system is deployed widely, a local design weakness can become a repeatable and scalable social problem. Ethical practice therefore requires technical validation, transparency, monitoring, and accountability throughout the system's life cycle.

References

- Drage, E., & Mackereth, K. (2022). Does AI debias recruitment? Race, gender, and AI's "eradication of difference." *Philosophy & Technology*, 35(1), Article 89. <https://doi.org/10.1007/s13347-022-00543-1>
- Electronic Privacy Information Center. (2019, November 6). *In re HireVue*. <https://epic.org/documents/in-re-hirevue/>
- Engler, A. (2021, March 12). Auditing employment algorithms for discrimination. *Brookings Institution*. <https://www.brookings.edu/articles/auditing-employment-algorithms-for-discrimination/>
- Fisher, S. L., Bonaccio, S., & Connelly, C. E. (2024). AI-based tools in selection: Considering the impact on applicants with disabilities. *Organizational Dynamics*, 53(1), Article 101036. <https://doi.org/10.1016/j.orgdyn.2024.101036>
- Harwell, D. (2019, November 6). A face-scanning algorithm increasingly decides whether you deserve the job. *The Washington Post*. <https://www.washingtonpost.com/technology/2019/10/22/ai-hiring-face-scanning-algorithm-increasingly-decides-whether-you-deserve-job/>
- Köchling, A., Riazzy, S., Wehner, M. C., & Simbeck, K. (2021). Highly accurate, but still discriminatory: A fairness evaluation of algorithmic video analysis in the recruitment context. *Business & Information Systems Engineering*, 63, 39–54. <https://doi.org/10.1007/s12599-020-00673-w>
- Köchling, A., & Wehner, M. C. (2020). Discriminated by an algorithm: A systematic review of discrimination and fairness by algorithmic decision-making in the context of HR recruitment and HR development. *Business Research*, 13, 795–848. <https://doi.org/10.1007/s40685-020-00134-w>
- Langer, M., König, C. J., & Papathanasiou, M. (2020). Highly automated interviews: Applicant reactions and the organizational context. *Journal of Managerial Psychology*, 35(4), 301–314. <https://doi.org/10.1108/JMP-09-2018-0402>
- Leicht-Deobald, U., Busch, T., Schank, C., Weibel, A., Schafheitle, S., Wildhaber, I., & Kasper, G. (2019). The challenges of algorithm-based HR decision-making for personal integrity. *Journal of Business Ethics*, 160(2), 377–392. <https://doi.org/10.1007/s10551-019-04204-w>
- Raghavan, M., Barocas, S., Kleinberg, J., & Levy, K. (2020). Mitigating bias in algorithmic hiring: Evaluating claims and practices. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency* (pp. 469–481). Association for Computing Machinery. <https://doi.org/10.1145/3351095.3372828>
- Sánchez-Monedero, J., Dencik, L., & Edwards, L. (2020). What does it mean to “solve” the problem of discrimination in hiring? Social, technical and legal perspectives from the UK on automated hiring systems. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency* (pp. 458–468). Association for Computing Machinery. <https://doi.org/10.1145/3351095.3372849>

- Selbst, A. D., Boyd, D., Friedler, S. A., Venkatasubramanian, S., & Vertesi, J. (2019). Fairness and abstraction in sociotechnical systems. In *Proceedings of the Conference on Fairness, Accountability, and Transparency* (pp. 59–68). Association for Computing Machinery. <https://doi.org/10.1145/3287560.3287598>
- Tabassi, E. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)* (NIST AI 100-1). National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.AI.100-1>
- U.S. Equal Employment Opportunity Commission. (2023, September 11). *iTutorGroup to pay \$365,000 to settle EEOC discriminatory hiring suit*. <https://www.eeoc.gov/newsroom/itutorgroup-pay-365000-settle-eeoc-discriminatory-hiring-suit>
- Wiessner, D. (2023, August 10). Tutoring firm settles U.S. agency’s first bias lawsuit involving AI software. *Reuters*. <https://www.reuters.com/legal/tutoring-firm-settles-us-agencys-first-bias-lawsuit-involving-ai-software-2023-08-10/>
- Woods, S. A., Ahmed, S., Nikolaou, I., Costa, A. C., & Anderson, N. R. (2020). Personnel selection in the digital age: A review of validity and applicant reactions, and future research challenges. *European Journal of Work and Organizational Psychology*, 29(1), 64–77. <https://doi.org/10.1080/1359432X.2019.1681401>
- Zuloaga, L. (2021, January 12). *Industry leadership: New audit results and decision on visual analysis*. HireVue. <https://www.hirevue.com/blog/hiring/industry-leadership-new-audit-results-and-decision-on-visual-analysis>